

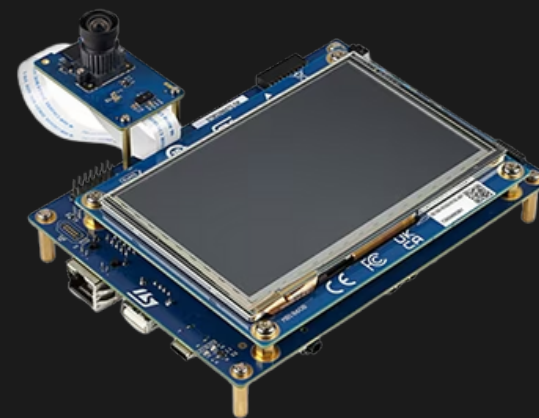
Demonstration of a Simple Transformer Running on the NPU of an STM32N6

ACA Project – A.Y. 2024/2025

Authors

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Supervisors

Prof. Silvano Cristina

Dott. Ronzani Marco



POLITECNICO
MILANO 1863





Objectives

TO BE ACHIEVED



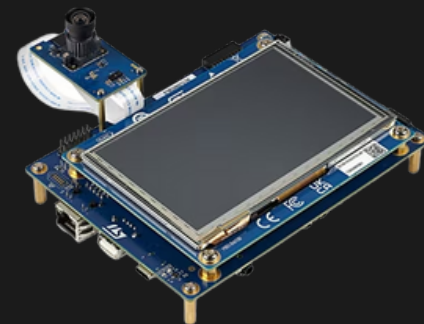
- 1
 - Deploy Model Zoo CNN
- 2
 - Build and Deploy Custom CNN
- 3
 - Full-Stack Transformer Deployment
- 4
 - Metric Collection & Architecture Comparison

Hardware

Toolchain

STM32N6570-DK

THE BOARD



LOCAL HARDWARE

2 NVIDIA GEFORCE RTX 5060 TI



ST Edge AI Developer Cloud

REMOTE BOARD



STM32CubeIDE

GENERATE CODE AND VALIDATE



STM32CubeProgrammer

FLASHING BINs



ANACONDA PROMPT

RUN PYTHON SCRIPTS



Task & Dataset

Supervised Image Classification: broad support in Model Zoo.

Flowers: 5 classes dataset

- 3.7K RGB images.
- 20% for quantization.
- 80/20 train-validation split.

Preprocessing: RGB conversion, resizing and normalization.

Augmentation: flip, contrast, brightness, translation, rotation, zoom.

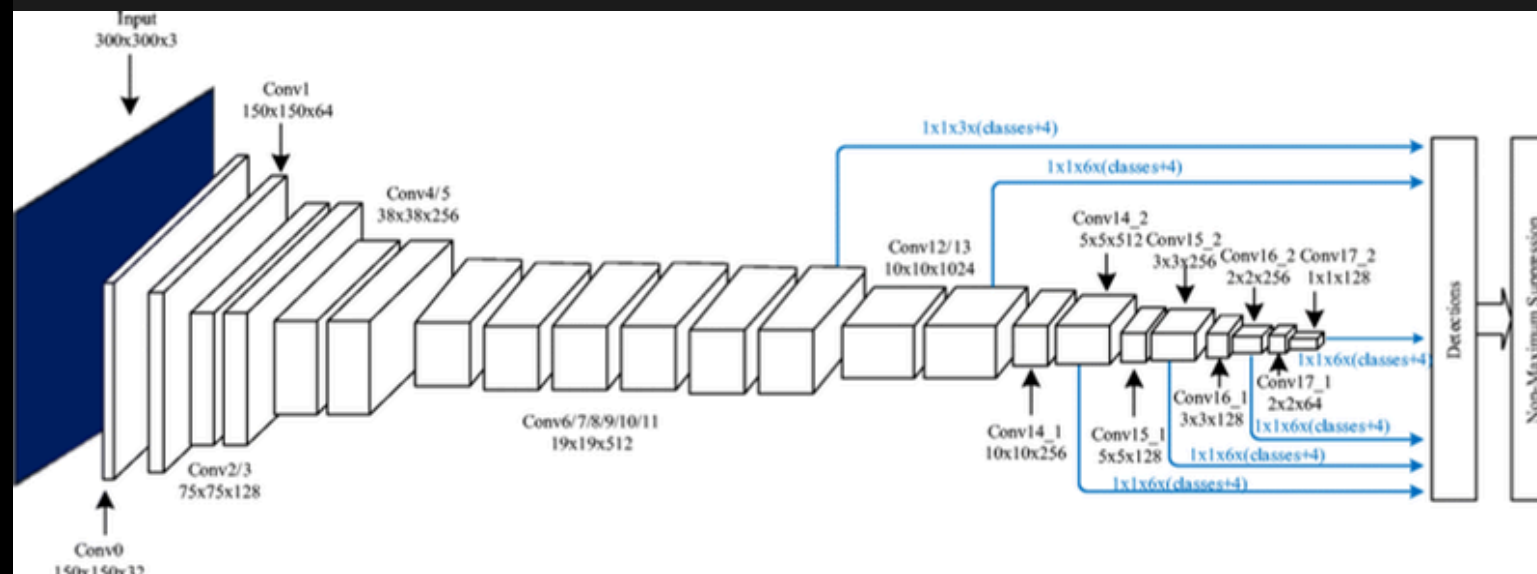


FLOWERS DATASET

Kaggle



CNNs



MOBILENET V2

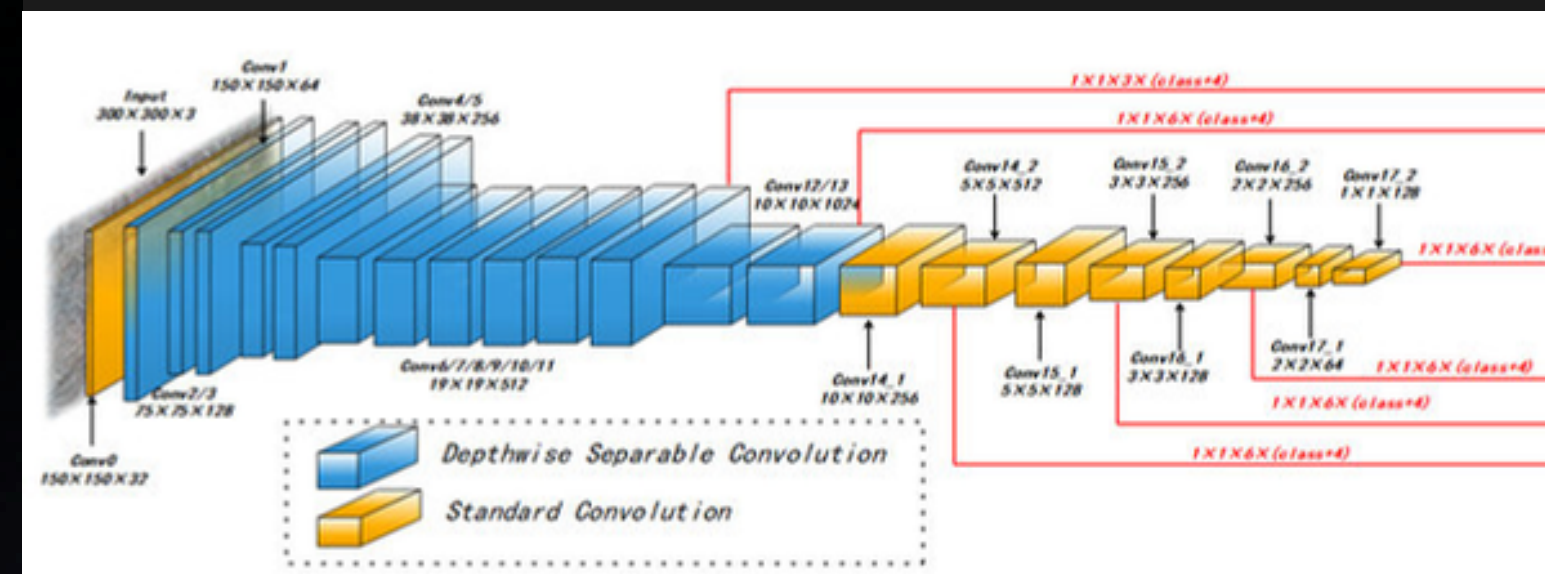
PRETRAINED ON IMAGENET

$\alpha = 0.35 \rightarrow \sim 16-1280$ CHANNELS

17 INVERTED RESIDUAL BLOCKS ($t=6$)

TRAIN: DROPOUT (0.3), 100 EPOCHS,
ADAM (lr=0.001)

QUANTIZATION: INT8 (POST-TRAIN)



CUSTOM CNNs

TRAINED FROM SCRATCH

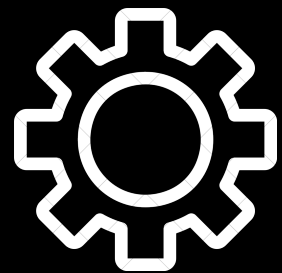
$\alpha = 1 \rightarrow \sim 32-128$ CHANNELS

4 INVERTED RESIDUAL BLOCKS ($t=2-3$)

TRAIN: DROPOUT (**0.25**), 100 EPOCHS,
ADAM (lr=0.001)

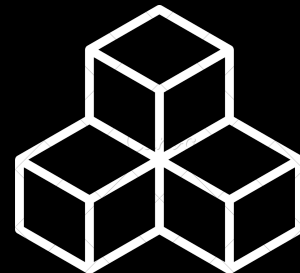
QUANTIZATION: INT8 (POST-TRAIN)

Transformer Architecture



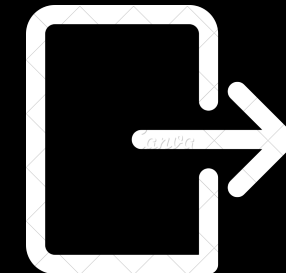
SPECS & INPUT

~4.56M parameters.
Input: 30 tokens.
Vocab size: 20K.
128-dim embeddings +
positional encodings.



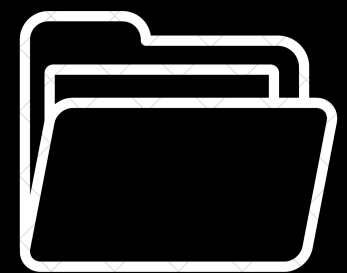
CORE STRUCTURE

6 encoder blocks.
4-head MHA.
FFN: Linear with ReLU.
Residual connections +
LayerNorm.



OUTPUT LAYER

Dense: 128 $\rightarrow$ 20K.
20K logits $\rightarrow$ softmax.



GENERATED FILES

Model weights: learned
during training.
Tokenizer: maps each
word to an index.
Embedding matrix: 20k x
128 size.

Tokenizer

CHARACTER-LEVEL

Vocab size: 66 (A–Z, a–z, digits, punctuation).

Embedding_size: $66 \times 128 \rightarrow$ very compact.

Generates 1 character per step → slow & verbose.

No PAD/OOV tokens → full coverage.

Low Flash/SRAM usage.

Poor semantic/syntactic learning.



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Tokenizer

WORD-LEVEL

Vocab size: 20'000 most frequent words.

Embedding size: $20K \times 128 \rightarrow$ large footprint.

Generates full words → efficient context usage.

OOV mapped to ID=1 → limited generalization.

High memory cost.

Better meaning retention, despite sparsity.



Datasets

Training Data

- Source: 4 classic English literary works.
- Merged using a custom Python script.
- Final size: 7.68 MB, ~1.4 million words.

Evaluation Data

- *The Adventures of Sherlock Holmes* by Arthur Conan Doyle.

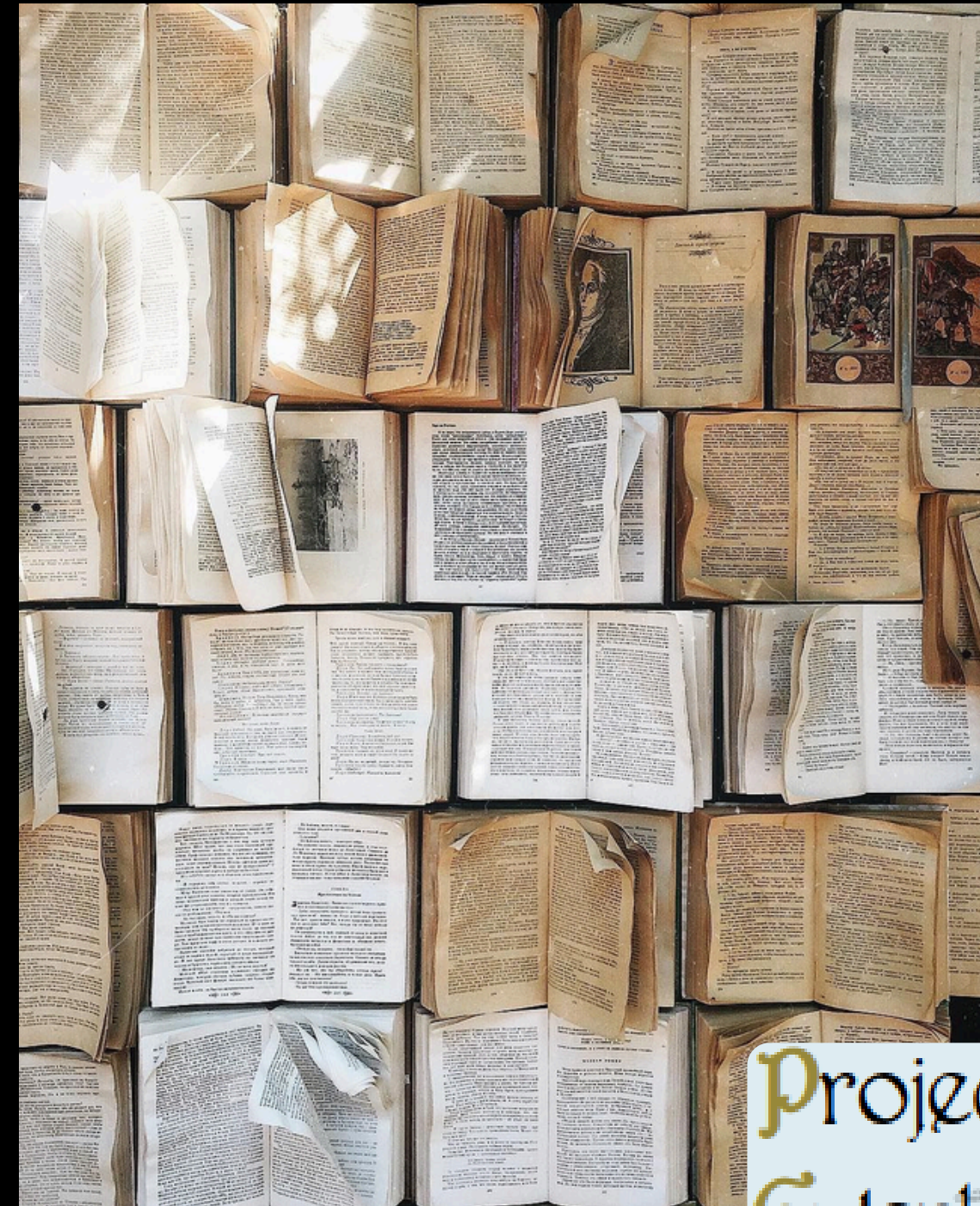
Reasoning

- Diverse, high-quality English.
- No overlap training-evaluation.
- Textual richness.



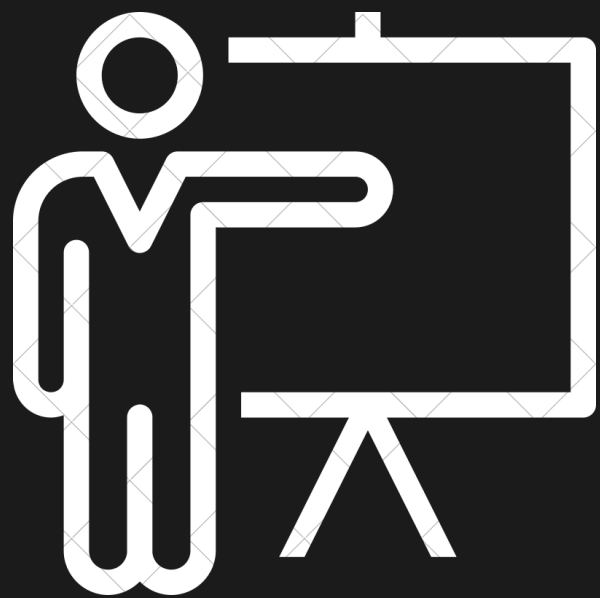
PROJECT GUTENBERG

75k EBooks



Project
Gutenberg

FULL STACK



TRAINING

90/10 TRAIN-VAL
100 EPOCHS



QUANTIZATION

INT-8 POST TRAIN &
CALIBRATION SET



life.augmented

DEPLOYMENT

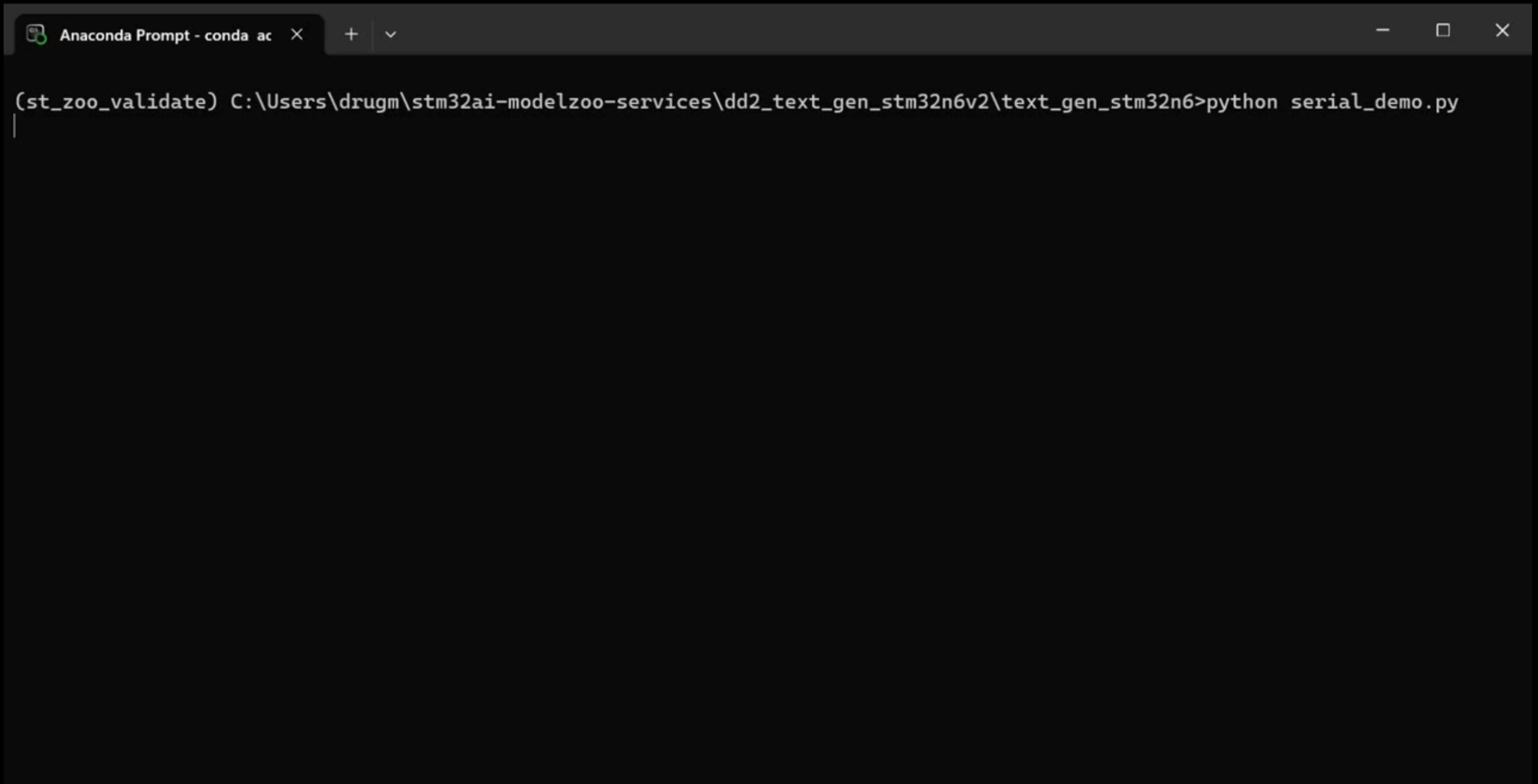
STM32CubeIDE
STM32Programmer



INFERENCE

AUTOREGRESSION
WITH NO MASKS

Now, let's admire this masterpiece of modern poetry.



```
Anaconda Prompt - conda ac  ×  +  ▾  
  
(st_zoo_validate) C:\Users\drugm\stm32ai-modelzoo-services\dd2_text_gen_stm32n6v2\text_gen_stm32n6>python serial_demo.py  
|
```


Model Metrics

Accuracy (14.8%) = 3000×
Perplexity (653) = 1/30



Embeddings

Cosine similarity between
pairs and single tokens

Query: You		
Rank	Token	Score
1	pause	0.7655
2	cat	0.7647
3	enough	0.7572
4	reality	0.7495
5	rumour	0.7474
6	spoke	0.7465
7	ginger	0.7443
8	brown	0.7433
9	examples	0.7370

On-Board

Model size & Tot Flash
Peak SRAM & Inference

Metric	Value	Unit / Notes
Flash footprint (model + embeddings)	7.159	MB (quantized, external Octo-SPI)
Model weights	4.590	MB
Int8 model parameters	4'562'979	values
Embedding file size	2.560	MB
Int8 embedding entries	20'000 · 128	values
Peak SRAM (activations + buffers)	1.208	MB
MACC	172,531,800	ops
Epochs executed	407	epochs
Epochs on NPU (hardware)	257	runs
Epochs on CPU (software)	150	runs
Tokenizer file size (host PC)	745	KB (JSON)
Generation length (reported run)	128	tokens
Time for 128-token generation	9.361	s (single representative run)
Average latency per generated token	73.14	ms/token (same run)
Throughput	13.67	tokens/s (same run)

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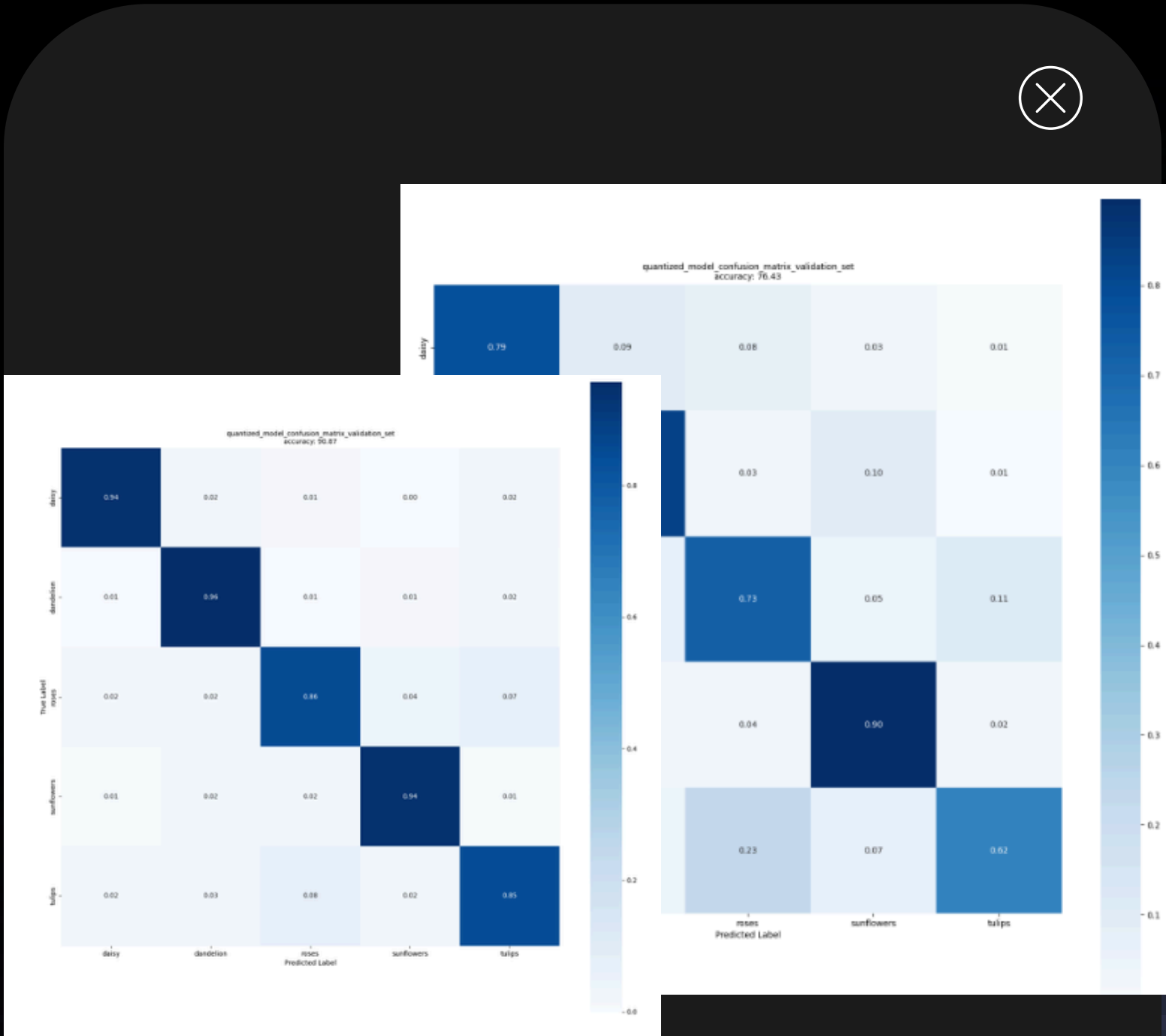
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On-Board

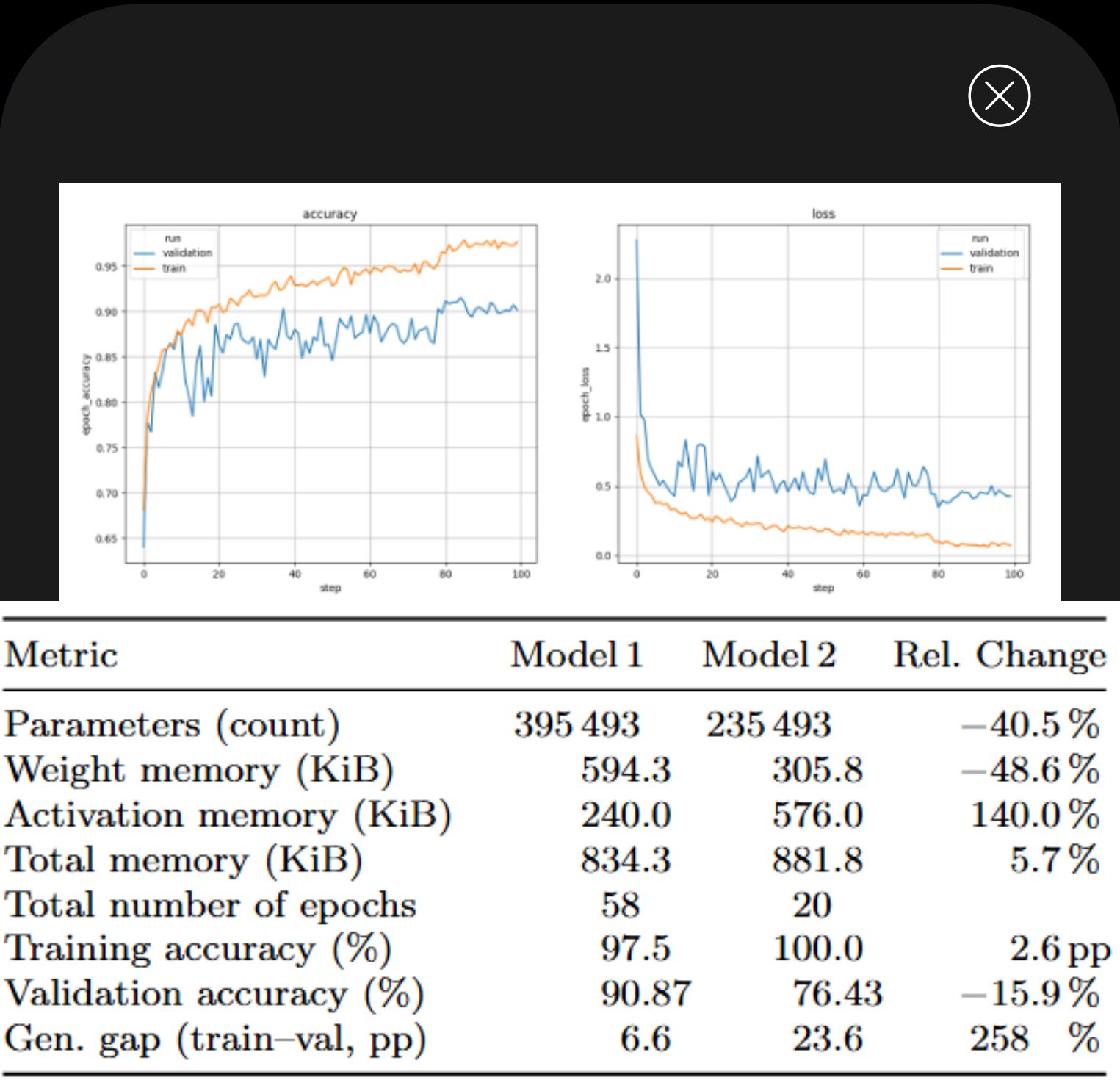
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CNNs



MODELZOO VS CUSTOM CNNs



MODELZOO VS CUSTOM CNNs

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5. Conclusions & Future Directions

Conclusions

- End-to-end INT8 Transformer (4.6 M params) successfully deployed on STM32N6570-DK.
- Fits comfortably on-board: ≈ 7 MB flash, ≈ 1.2 MB SRAM.
- 73 ms per token $\rightarrow$ 13.7 tokens/s; NPU handles 63% of epochs.
- 14.8% accuracy & 653 perplexity on unseen text. $\sim 3\,000\times$ above random uniform baseline.
- Embeddings show some meaningful word-level semantics, validating architecture choice.
- Besides educational purposes, there are only a few practical applications for an embedded Transformer for text generation.

Future Work

- Train on larger and cleaner datasets.
- Explore language adaptation to Italian.
- Expand embedding size (128 $\rightarrow$ 256).
- Fine-tune larger models (deeper/larger blocks) using spare flash/SRAM.



Thank you

QUESTIONS?

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MILANO 1863



Query: Romeo		
Rank	Token	Score
1	thee	0.7894
2	withdrew	0.7832
3	dice	0.7825
4	interview	0.7820
5	consists	0.7819
6	stricken	0.7808
7	roared,	0.7792
8	suspicious	0.7784
9	yield,	0.7768
10	witness	0.7746

Query: You		
Rank	Token	Score
1	pause	0.7655
2	eat	0.7647
3	enough	0.7572
4	reality	0.7495
5	rumour	0.7474
6	spoke	0.7465
7	ginger	0.7443
8	brown	0.7433
9	examples	0.7370
10	commence	0.7355

Query: King		
Rank	Token	Score
1	Ephesus.	0.7676
2	conversed	0.7667
3	well.	0.7612
4	death's	0.7572
5	marry	0.7566
6	gravel	0.7480
7	men's	0.7480
8	sooner	0.7460
9	Master,	0.7458
10	Either	0.7450

Cosine(Romeo, Juliet)	0.6538
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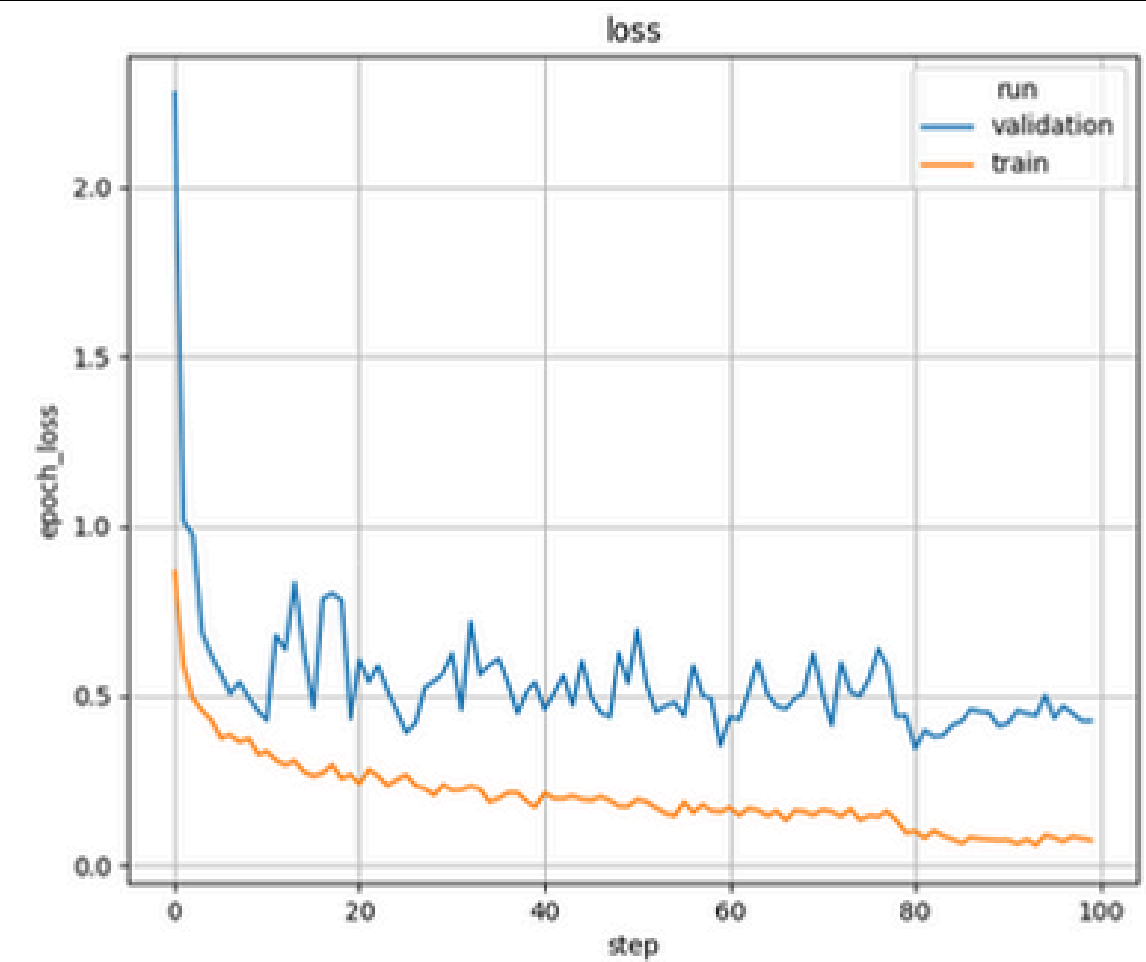
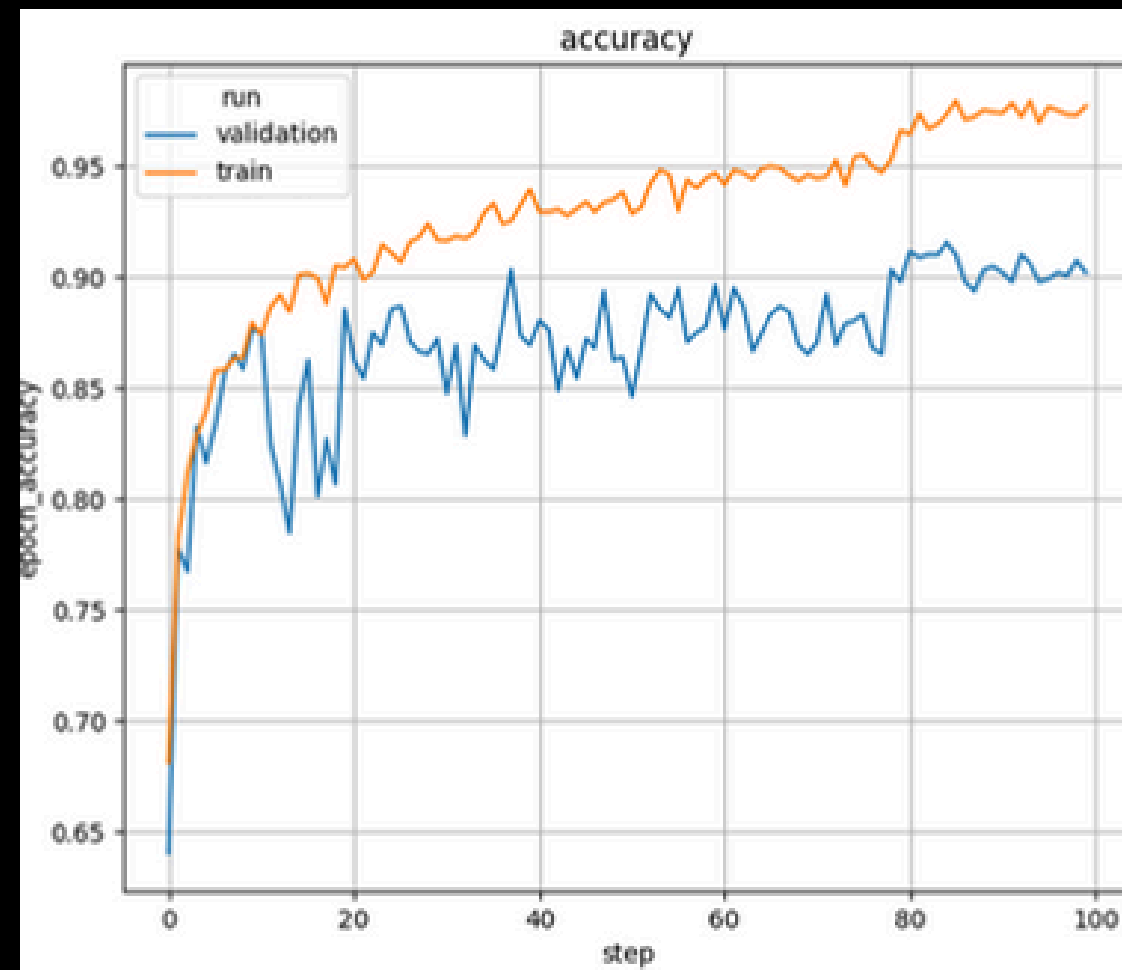
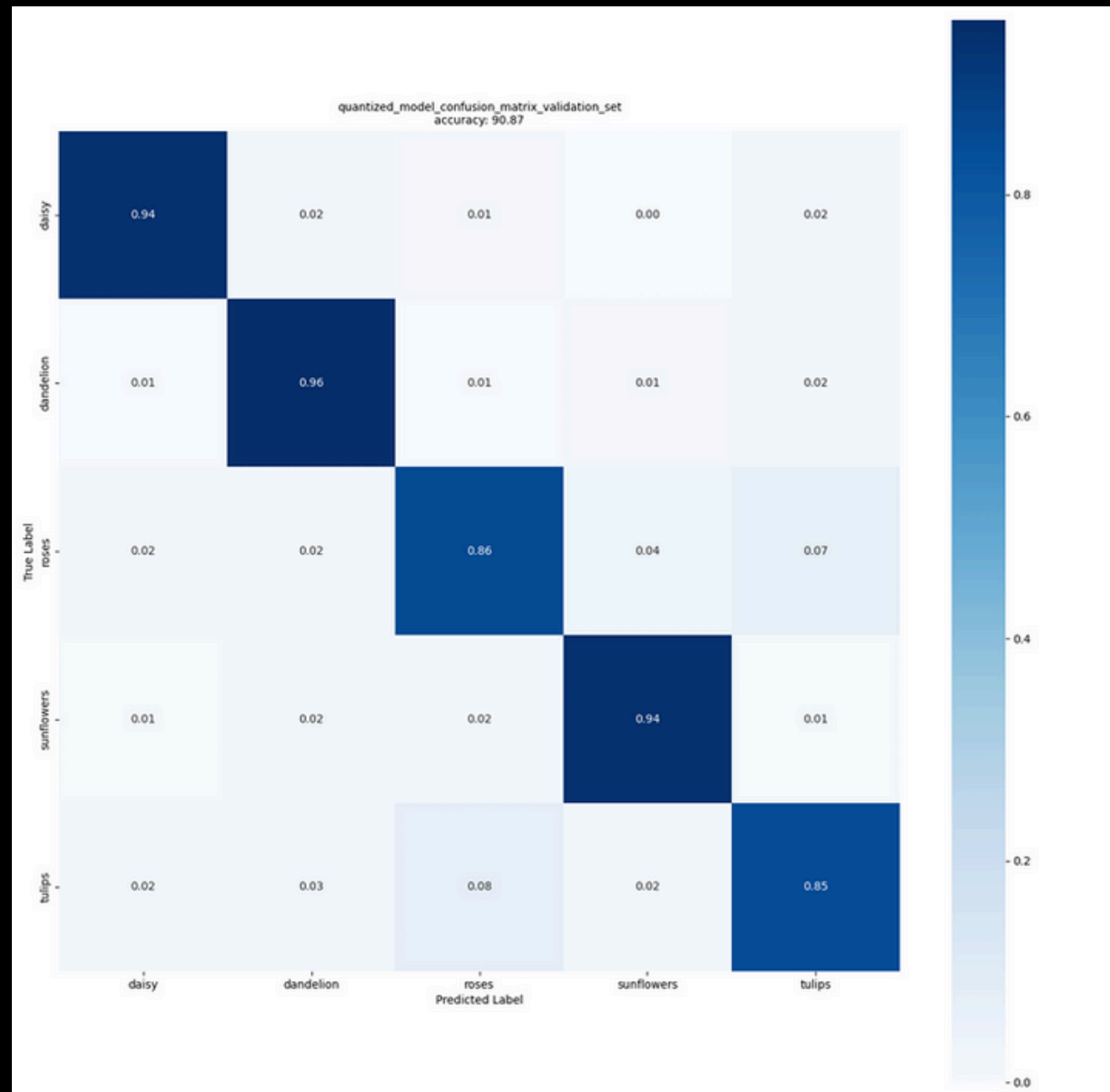
Table 4: Top-10 most similar tokens for each query, plus the cosine similarity between *Romeo* and *Juliet*.

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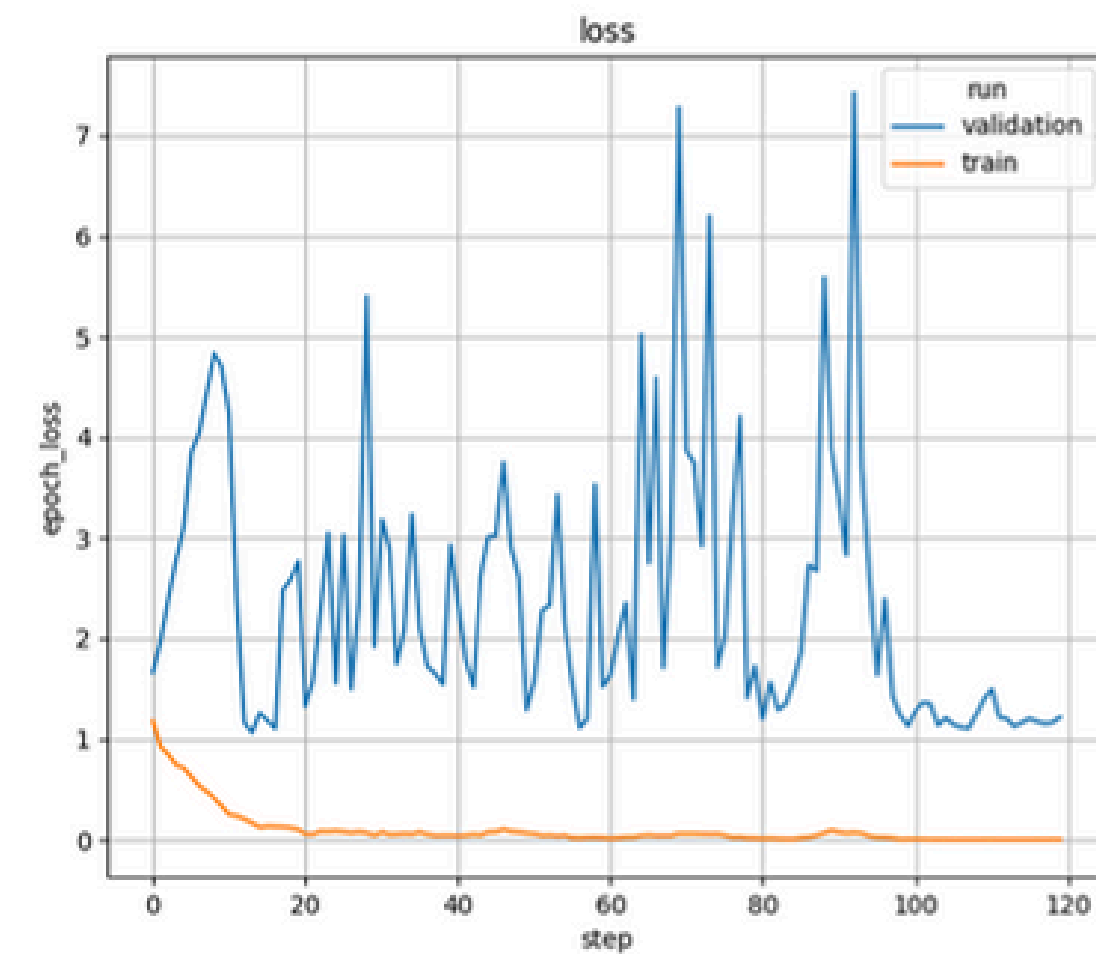
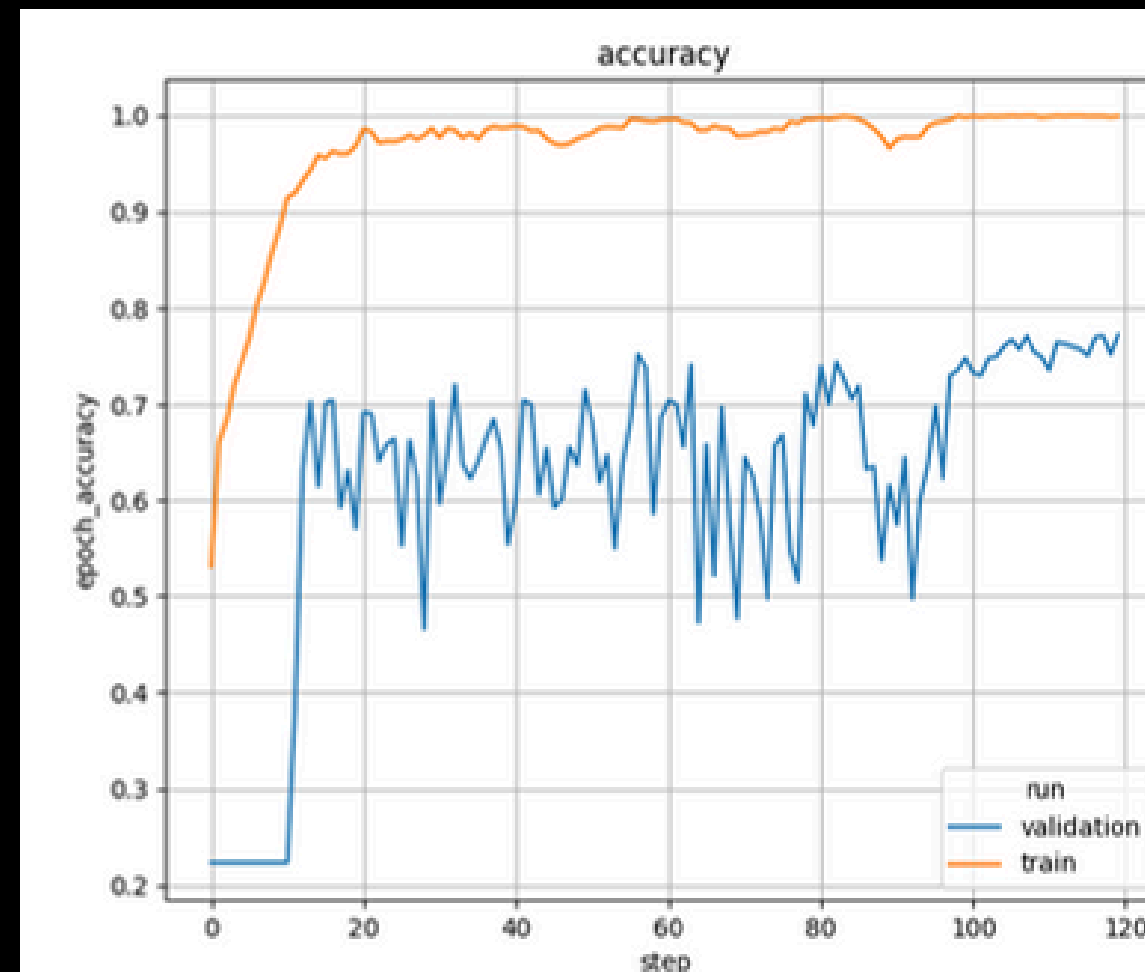
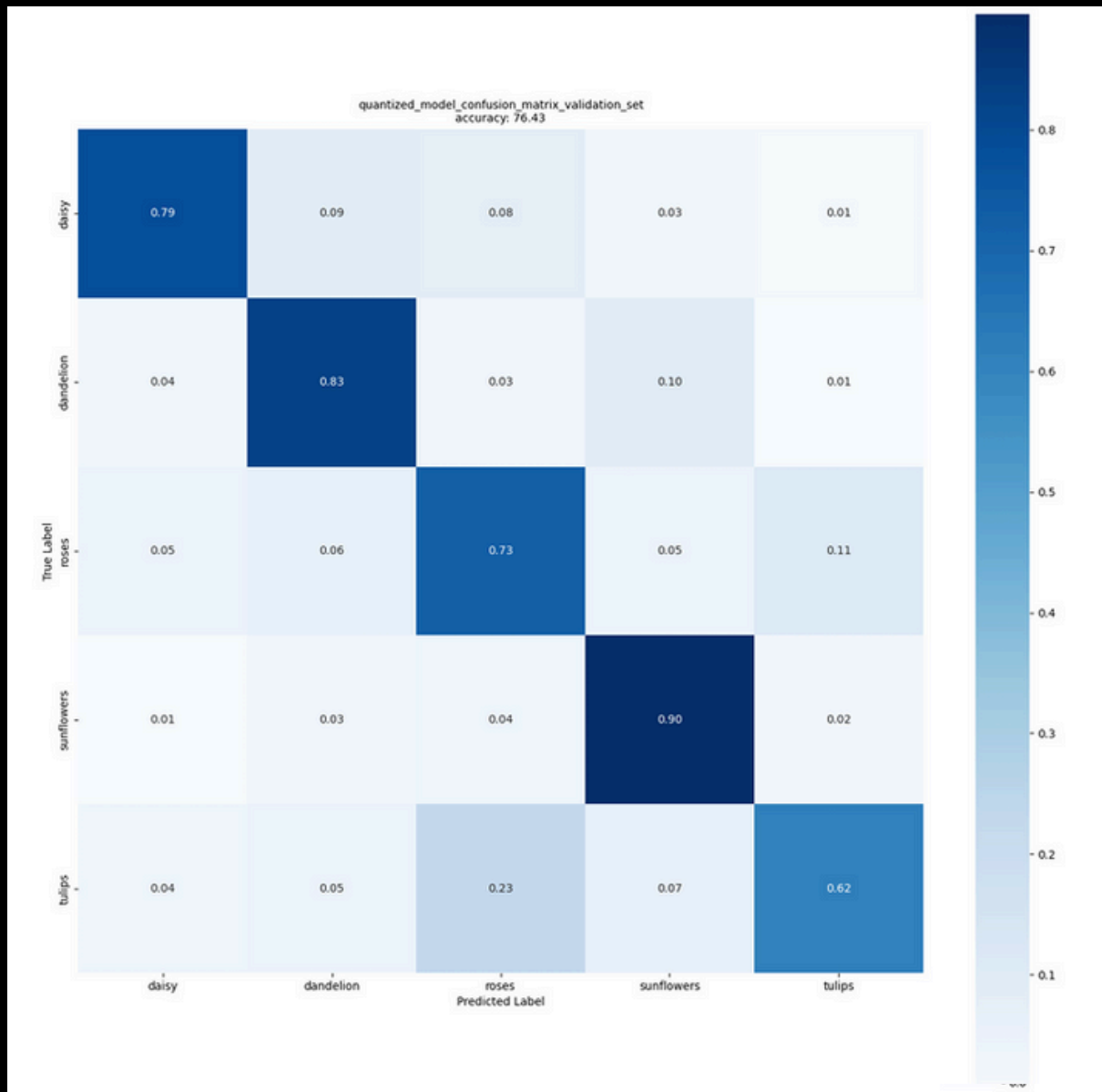
Region	Address range	Usage			Weights	Activations
		Used	Total	% Used		
flexMEM	0x34000000–0x34000000	0 B	0 B	0.00%	0 B	0 B
cpuRAM1	0x34064000–0x34064000	0 B	0 B	0.00%	0 B	0 B
cpuRAM2	0x34100000–0x34200000	1.000 MB	1.000 MB	100.00%	0 B	1.000 MB
npuRAM3	0x34200000–0x34270000	147.875 kB	448.000 kB	33.01%	0 B	147.875 kB
npuRAM4	0x34270000–0x342E0000	0 B	448.000 kB	0.00%	0 B	0 B
npuRAM5	0x342E0000–0x34350000	64.688 kB	448.000 kB	14.44%	0 B	64.688 kB
npuRAM6	0x34350000–0x343C0000	0 B	448.000 kB	0.00%	0 B	0 B
octoFlash	0x71000000–0x78000000	7.159 MB	112.000 MB	6.40%	7.159 MB	0 B
hyperRAM	0x90000000–0x92000000	0 B	32.000 MB	0.00%	0 B	0 B
Total		8.367 MB			7.159 MB	1.208 MB

Table 2: Memory usage information (including embedding file; I/O buffers are included in activations)

ModelZoo CNN



Custom CNN



Metric	Model 1	Model 2	Rel. Change
Parameters (count)	395 493	235 493	−40.5 %
Weight memory (KiB)	594.3	305.8	−48.6 %
Activation memory (KiB)	240.0	576.0	140.0 %
Total memory (KiB)	834.3	881.8	5.7 %
Total number of epochs	58	20	
Training accuracy (%)	97.5	100.0	2.6 pp
Validation accuracy (%)	90.87	76.43	−15.9 %
Gen. gap (train–val, pp)	6.6	23.6	258 %